

3 Gelatine is a protein derived from collagen.

The photograph shows a packet of gelatine capsules containing medicine.

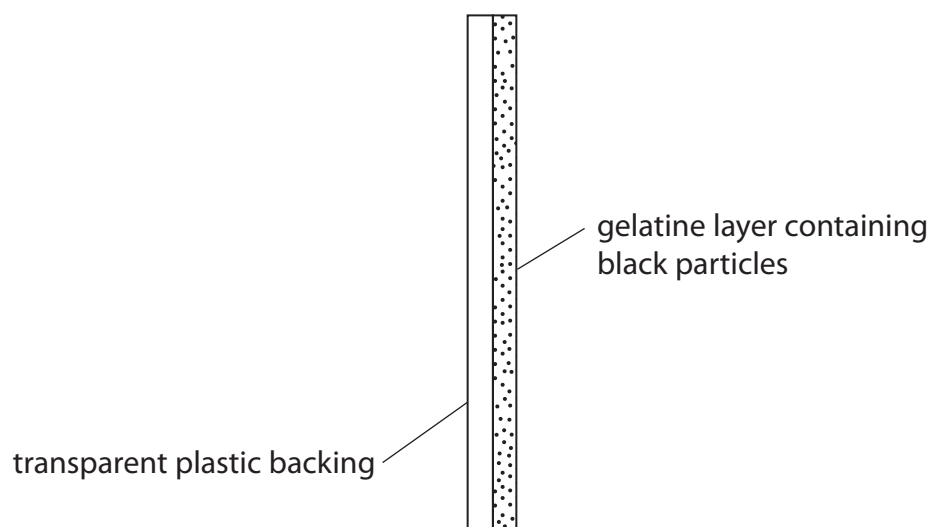


(Source: © PAL)

In the stomach, the capsule is broken down by protease to release the medicine.

The effect of protease concentration on the digestion of gelatine was investigated.

Pieces of photographic film, which have a layer of gelatine containing black particles, were used. As the gelatine is digested, the black particles fall off and the film becomes transparent.



Protease activity was measured by timing how long it took the film to become transparent.

(a) (i) State the dependent variable in this investigation.

(1)

(ii) One of the variables that needs to be controlled in this investigation is pH.

Describe how a suitable pH could be decided on experimentally.

(3)

(b) The results of this investigation are shown in the table.

The proportional rate of reaction = $\frac{1}{\text{time}}$

Protease concentration (%)	Time taken for film to become clear / seconds	Proportional rate of reaction / seconds ⁻¹
1.0	449	0.0022
1.5	288	0.0035
2.0	252	0.0040
3.0	242	0.0041
4.0	230	0.0044
5.0	236	0.0042



- [illegible]

- (4)

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

- (c) (i) This investigation was criticised because it does not measure the initial rate of reaction.

Explain why it would be better to measure the initial rate of reaction.

(2)

.....

.....

.....

.....

.....

.....



- (ii) Suggest how the initial rate of reaction could be measured in this investigation.

(3)

- (d) A student repeated the result for the 2% protease solution.

Determine how 1 cm³ of a 2% solution could be obtained from a 5% stock solution of protease.

(2)

(Total for Question 3 = 19 marks)

TOTAL FOR PAPER = 50 MARKS

